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Let $e_1 = \begin{bmatrix} 1 \\ \vdots \\ 0 \end{bmatrix}$ $e_2 = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$ $\cdots$ $e_n = \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$ and $u = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$

$\Rightarrow$ We can express u in terms of : $u = c_1e_1 + c_2e_2 + c_3e_3 \cdots c_ne_n$

$\Rightarrow T(u) = T(c_1e_1 + c_2e_2 \cdots + c_ne_n) = c_1T(e_1) + c_2T(e_2) \cdots + c_nT(e_n) = (T(e_1) + T(e_2) + T(e_3) \cdots) \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_n \end{bmatrix}$

Thus transformation $T(\ )$ of matrix is A, $A = [T(e_1), T(e_2) \cdots T(e_n)]$

A is also called standard matrix.